Assessment – Week 3 – Data Analysis with Python

Final project: I would like to form a group with Dimitri Siegrist.

Completed tasks:

- Because I feel quite confident in Python fundamentals, I skipped the one-hour video to concentrate on the other tasks.
- I started the homework by reading the "Tidy Data" paper by Wickham. I tried taking some notes to remember the most important things.
- Afterwards, I started with the intro to Pandas. Firstly, I looked at the "10min tutorial for a quick start". This felt a little bit confusing to me, which is why I switched to the Getting started tutorials. These were a lot easier to understand. I have worked in R already and, to my surprise, there are some similarities.
- For the practice part, I chose the Disney dataset. I basically worked through the five tasks. I analysed whether the release season of a Disney movie is related to its vote-weighted IMDb rating. The weighted mean rating was lowest for winter releases (7.17) and similar for spring, summer and autumn (7.47 to 7.52), so the differences are small and I did not yet test whether they are significant. The practice part took way longer than expected. It still helped a lot to understand Pandas further (just by reading, it is harder for me to remember things).
- During the week, I also brainstormed about the project. Our first idea is to build a Streamlit app ("Snow Guru") about skiing and snow, in general, in Switzerland. As I love skiing on the slopes and Dimi does ski touring, we would like to build some features to support our hobbies. Two of the features might be a snow melt model for ski touring and a comparison of ski resorts, giving the user a ranking of which ski resort will have the highest "consumer surplus" (we will take variables like day pass cost, snowfall, slope kms, distance to ski resort etc. into account).

What to work on in the next weeks/what I struggle with:

- This was by far the most challenging homework so far, as I had never worked with Pandas before. Therefore, the homework took much longer than in the previous weeks.
- The hardest part was trying to understand Pandas. In the following weeks, I would like to get a bit more fond of the syntax. Hopefully, this will get easier with time.
- I also underestimated coming up with a project idea that is a good mix between interesting and not too ambitious. It is quite hard to evaluate which ideas would be too hard for us to implement.

GitHub repo: https://github.com/luisasnbl/Homework_Data_Analysis_Python.git